

Scenario_BiodivChanges

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Basic run

The model is initialized without any change to the input parameter, but the spatial window (North America).

```
library(terra)
library(MadingleyR)
spatial_window= c(-170, -50, 12, 85)
plot_spatialwindow(spatial_window)
sptl_inp <- madingley_inputs("spatial inputs")
chrt_def = madingley_inputs("cohort definition")
stck_def = madingley_inputs("stock definition")
mdl_prms = madingley_inputs("model parameters")
mdata = madingley_init(spatial_window = spatial_window, cohort_def = chrt_def, stock_def=stck_def, spatial_i
```

Then, we run the model for an *initial period* of **100 years** and a *max cohort size* of **100** to let it stabilize before the scenarios

```
mdata2 = madingley_run(madingley_data=mdata, years=100, cohort_def=chrt_def, stock_def=stck_def, spatial_i
```

From this point, we use this initial run to implement our different scenarios.

Create a reference scenario

To compare the results, we let the model run for the same amount of time the scenarios would run (ADD NUMBER AFTER NORMALIZING SCENARIOS)

```
mdata3 = madingley_run(madingley_data=mdata2, years=50, cohort_def=chrt_def, spatial_inputs = sptl_inp)
```

Scenario: Climate change

The first scenario looks into the impacts of projected climate change on biodiversity. The input data were gathered from CHELSA CMIP6 (SSP585, 2050s) climate inputs (tmax/tmin → tmean) and aggregated from 30 to 1° for MadingleyR compatibility. Datas were resampled to a **monthly temporal resolution**, and a *spatial resolution* of **1 degree**. This resulted in (NUMBER), 12 per year from 2041 and 2070.

Scenario: HANPP

To represent projected human impacts on biodiversity, we used the projected land-use under socio-climatic scenarios from Chen, Li & Liu (<https://www.nature.com/articles/s41597-022-01208-6>). We used data from the SSP5_RCP85 to match the datasets of the Climate change scenarios (see above). The dataset represents land-use in **5 years** intervals, from **2020** to **2100**.

Modifications were made to the dataset to make them MadingleyR-compliant. These were mostly done using QGIS 3.40.7, and only the resulting data are provided for size reasons. # A QGIS Model is available to fully batch-reproduce these changes. Modifications: - Reprojection: Make sure every raster is in the same projection (4326) - Clip raster to spatial window: From a mask layer representing the same spatial window, to reduce the size of the files for better handling. Could be skipped to use the same data worldwide. - Convert land-use data to HANPP. Following Haberl 2007 (<https://www.pnas.org/doi/epdf/10.1073/pnas.0704243104>), we converted the land-use values to HANPP equivalent values. (VALIDATE MODEL, PROBABLY HAVE TO INVERT) Anthropogenic presence: 0.73 Agricultural land: 0.835 Natural habitat: 1 Everything else: 0 - Resample: The initial data used a 1km resolution, while everything else in MadingleyR used a 1 degree resolution. The dataset was resampled to 1 degree. - Fill the raster To prevent any issue, the output raster was formatted to span across the whole world, filling the NULL data with 0.

```
# Get Karen's code here
```

Scenario: Remove large herbivores

To represent extinction events, we removed certain cohorts. In this case, all the large herbivores were removed. We reproduced the tests made by (<https://madingleyr.github.io/MadingleyR/CaseStudies/CASESTUDY1.html>).

```
# Get Simran's code here
```

Scenario: Remove large carnivores

To represent extinction events, we removed certain cohorts. In this case, all the large carnivores were removed. This test is inspired by (<https://madingleyr.github.io/MadingleyR/CaseStudies/CASESTUDY1.html>).

```
mdata_carni = mdata2

remove_idx = which(mdata_carni$cohorts$AdultMass > 1e5 & mdata_carni$cohorts$FunctionalGroupIndex == 1)

cohort_defs$PROPERTY_Maximum.mass[2] = 100000

cohort_defs=madingley_inputs('cohort definition')
```

Scenario: Climate change+HANPP